Identifying the Greatest Tour de France Riders Through Network Science

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In this project, we present a novel application of network analysis to evaluate over a century of Tour de France results (1903–2024) in order to rank the most dominant riders in the event's history. We construct a series of directed multi-digraphs, where each rider is connected to those who finished ahead of them in individual stages, forming stage-level hierarchies. Aggregating these connections across all stages and editions produces large-scale competitive networks that capture the dynamics of rider performance over time. We explore six network variants using different edge-weighting strategies, including time differences, point differentials, and unweighted rankings, each offering a distinct lens into rider consistency and dominance. Applying the PageRank algorithm to these networks allows us to rank cyclists not just by victories, but by the quality of their performances against strong competitors. This approach provides a more nuanced, data-driven framework for historical comparison, enabling recognition of not only overall winners but also consistently high-performing specialists across different race conditions and eras.

Introduction The Tour de France is widely regarded as the most prestigious cycling event in the world. This multi-stage competition challenges elite cyclists across a variety of terrains—from flat sprints to grueling mountain climbs—testing both endurance and strategic skill. The race boasts a rich history of legendary athletes, from the universally acclaimed all-time great Eddy Merckx to emerging stars like Tadej Pogačar.

Comparing cyclists across different eras, however, presents a significant challenge. Variations in competition formats, technological advancements, and changing race strategies make direct performance comparisons difficult. Traditional methods of comparison often rely on basic statistics such as the number of stage wins, podium finishes, or overall victories. While informative, these metrics can overlook the nuanced contributions of different rider types—such as climbers, sprinters, and time-trial specialists.

Another possible approach involves analyzing physical performance indicators like average power output or speed. However, these metrics tend to favor modern cyclists who benefit from vastly superior equipment, training methods, and team support structures. As a result, such comparisons may not accurately reflect a rider's skill or impact within their competitive context.

To address these limitations, we propose a novel approach using network analysis to evaluate and compare Tour de France cyclists. By modeling rider interactions and relative performance across stages and editions of the race, we aim to produce a more context-sensitive and data-driven ranking that captures the complexity and diversity of rider achievements over more than a century of competition.

Related work

The application of network analysis in sports research is vast; it can be used to uncover team dynamics, player interactions and performance metrics across various sport disciplines and much more. For instance, Bi et al. (1) applied network analysis to examine how socio-demographic factors such as age, gender, and education influence sports participation in China. In team sports such as football and basketball, passes are a crucial aspect of the game. The works by Pena & Touchette (2) as well as Fewell et. al. (3) explore the importance of passing by constructing passing networks and analyzing their structural properties and uncovering unique team strategies.

The PageRank algorithm (4), originally developed to rank web pages, has been successfully adapted to the realm of sports analytics. Several researchers have demonstrated its utility as a robust tool for ranking players and predicting match outcomes. For example, Matthews et al. (5) employed a weighted PageRank approach for NCAA basketball league to rank basketball teams, incorporating factors such as score margin, date, and venue, and demonstrated that it could outperform traditional seeding methods for who may participate in the competition. Shi and Tian (6) proposed a generalized PageRank model that integrates Bayesian correction with game outcomes to enhance the prediction accuracy of team rankings in the NBA. On a different note, Mukherjee (7) used PageRank together with centrality measures to analyze the performance of cricket players, constructing a directed and weighted network based on individual match interactions.

Beyond team sports, network analysis has also been effectively applied to individual sports. In tennis, Radicchi (8) applied PageRank to head-to-head match data to identify top players in history, demonstrating the algorithm's strength in individual, tournament-based sports. In Formula One, Horvat et al. (9) modeled driver performance using directed multigraphs, where drivers are nodes and finishing positions are represented as edges. These networks allow for the application of centrality and community detection algorithms, including PageRank, to identify dominant drivers and race patterns across seasons. In this project, we adopt and extend Horvat's methodology to the Tour de France by considering different kind of constructed networks and testing the scalability of their approach on a much larger network.

In terms of cycling, no major studies delve into ranking the historic achievements of cyclists. However there are various studies that focus on the prediction of upcoming race winners presented by Kholkine et al. (10) and the prediction of upcoming talent for example in the work by (11) but none of

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those studies use network analysis approaches.

Results

We generated six distinct Tour de France rider networks based on different edge-weighting schemes and analyzed them using PageRank and community detection. The Fruchterman-Reingold layout was used to visualize the rider networks, with Leiden clustering identifying communities in each. Leiden grouped bikers by generations. In visualization we only visualized top 2000 nodes (by PageRank) and labeled only best 30 of them, while increasing the size of performant riders and decreasing opacity for edges with lesser weight.

A. Scaled Time Difference Network. In the Scaled time difference network (Figure 1), mountain stages and time trials are deemed as more important. In this network we observe riders that typically perform well in general classification, thus is a good approximation of the statistics table results.

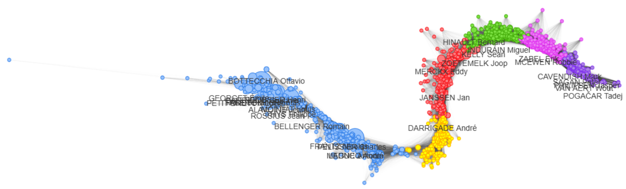

Fig. 1. Scaled Time Difference Network

B. Pure Points Network. Pure Points network (Figure 2) creates a network focused purely on point-scoring performances. Riders who often place just outside point positions are excluded. It's best for identifying sprinters and GC contenders alike.

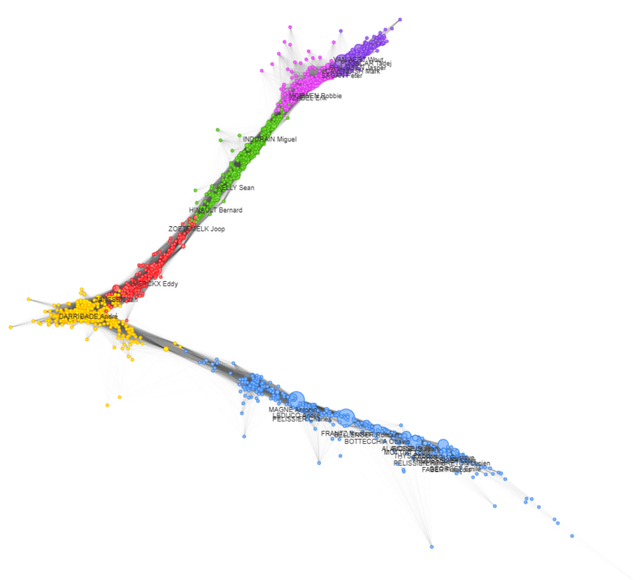

Fig. 2. Pure Points Network

C. Points Network. In Points network (Figure 3), the scores are allocated similarly to official rankings. So, a cyclist with more races will rank higher than a cyclist with less races, because he will have more points accumulated in his career so far.

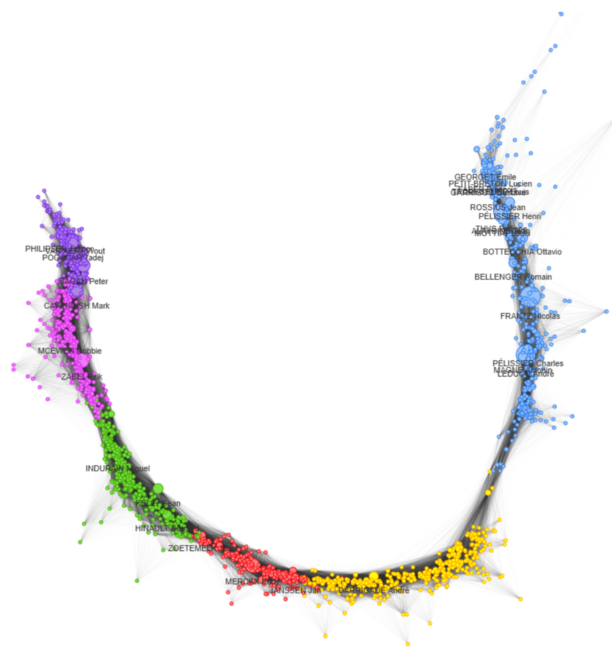

Fig. 3. Points Network

D. Unweighted Network. In the unweighted network (Figure 4), all weights equal to value of 1. This means that the network gives higher scores to cyclists that are also rated highest on official rankings. It shows how consistent they were throughout their career. High PageRank means that the person was winning a lot (did not have many people who finished before him).

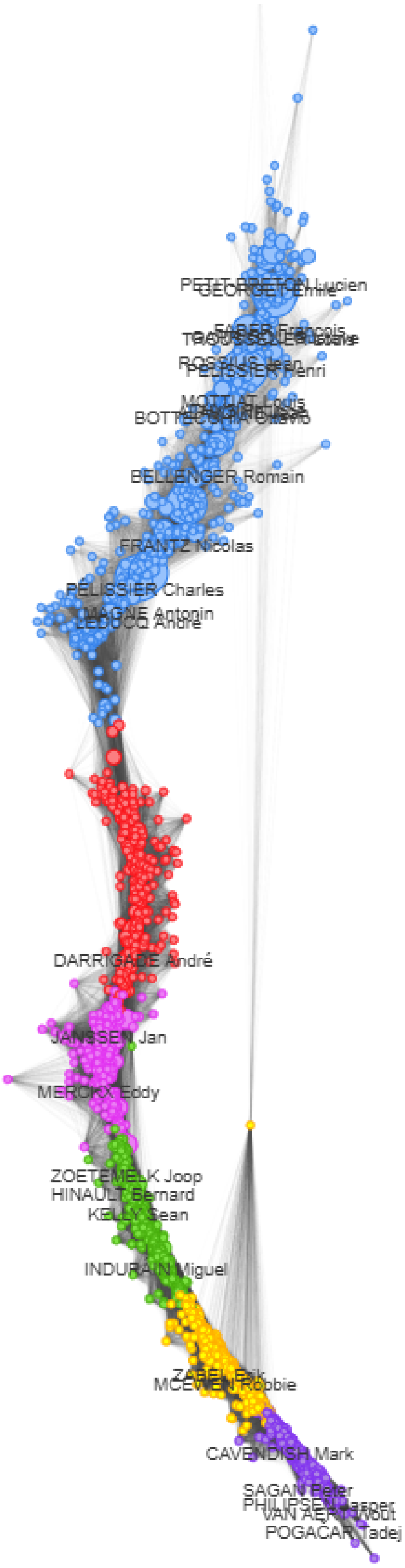

Fig. 4. Unweighted Network

E. Normalized Time Difference Network. Normalized time difference network (Figure 5) is similar to the Time Difference Network, but the times are normalized. So for example 10 seconds in a shorter race means bigger dominance than 10 seconds in longer race. Doing that, we regularize different lengths of the stages.

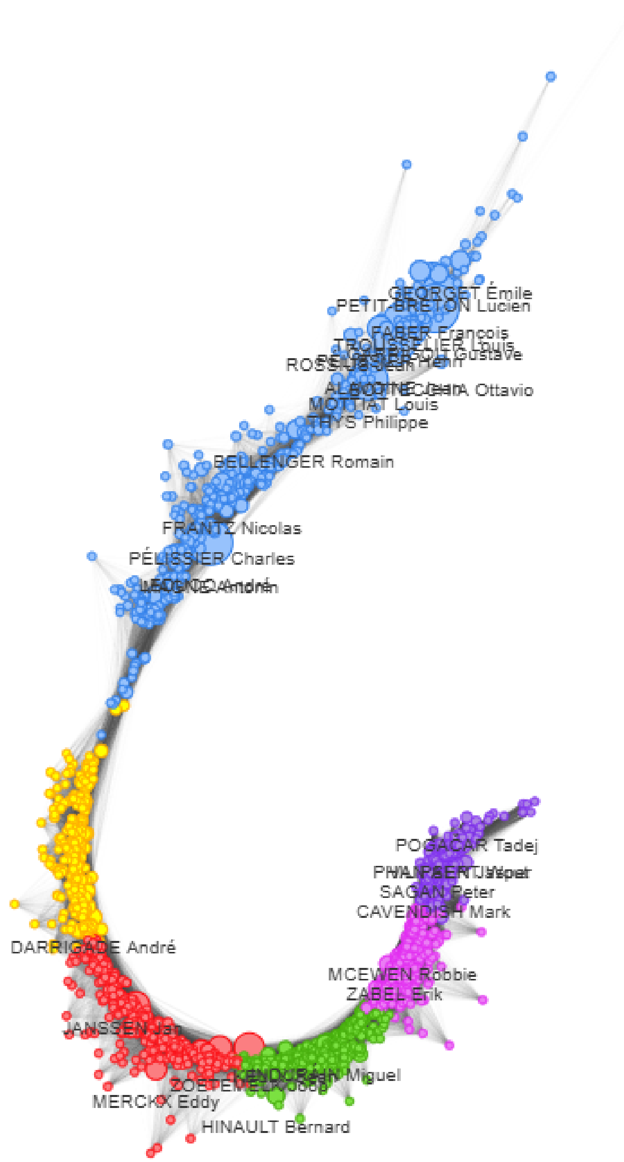

Fig. 5. Normalized Time Difference Network

F. Time Difference Network. Time Difference network (Figure 6) tells us how (non)dominant the cyclist was. The more time difference there was between the cyclist and others, the higher score the cyclist got. So if one person participated in small amount of races, but he dominated greatly, he will be placed higher than someone who won more times but with smaller time margin.

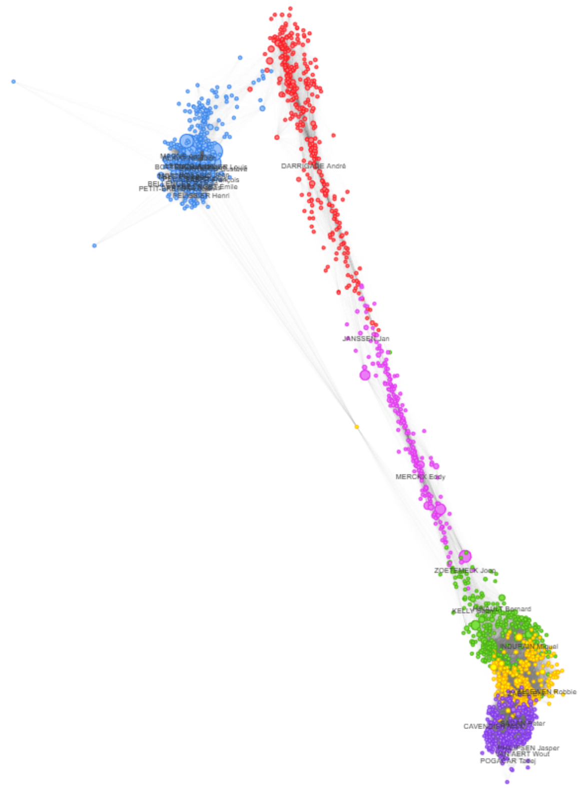

Fig. 6. Time Difference Network

Methods

Data - To carry out our project, we collected data on Tour de France races from the website ProCyclingStats.com using a custom web scraping script. This allowed us to extract multiple types of rider networks as well as a general statistics table. We constructed each network in parallel, updating them stage by stage alongside the statistics table. To define the edge weights of each network, we employed the following approaches:

- **Unweighted:** Edges are unweighted, riders point toward riders placed ahead of them.
- **Points:** Edges are weighted based on the difference in points scored between riders.
- **Pure Points:** Similar to the Points network, but only riders who scored points in the stage are included.
- **Time Difference:** Edges are weighted according to the absolute time difference between riders at each stage.
- **Normalized Time Difference:** Time differences are normalized, allowing comparison across stages with different time ranges.
- **Scale Time Difference:** Time differences are scaled to emphasize specific types of stages (e.g., mountain stages, time trials, flat stages), assigning greater importance to more selective or decisive stages.

Due to the extensive historical range of Tour de France results (dating back to 1903), we encountered several inconsistencies in the finishing times on ProCyclingStats.com. In cases of conflicting data across stages, we resolved issues by copying the finishing time of the preceding rider, ensuring increasing order of finishing times. The statistics table enables quick access to key summary information such as the number of top-5 finishes per rider and total jersey wins across all Tours.

Methods - We applied the standard PageRank algorithm to each constructed network to rank riders based on their centrality within the graph. To visualize the network structure and interpret the results, we used the Fruchterman–Reingold force-directed layout (12). This visualization helped identify the types of riders emphasized by PageRank across different network definitions.

To maintain clarity in the visual representation, we limited our graphs to the top 2,000 riders, with node size scaled according to PageRank score and only the top 30 riders explicitly highlighted while also coloring different generations of riders with the help of Leiden (13) clustering algorithm.

For comparative analysis, we also explored combining the PageRank scores from different network types by summing the scores, yielding a composite ranking that reflects multiple performance dimensions.

Discussion

This project presents a network-based analysis of Tour de France rider performance from 1903 to 2024. We constructed six different networks of riders using a range of edge-weighting strategies (time-based, point-based, and unweighted). With this, we tried to get different aspects of rider dominance and consistency across the races and tours. We used the PageRank algorithm to identify best-performing riders and not just overall winners. The Fruchterman-Reingold layout provided a visualisation of the rider's performance over time.

Each of our six networks provided a unique look at the riders:

- **Scale Time Difference:** Find's dominance in flat stages where gaps are minimal
- **Pure Points:** Identifies the stage winners or consistent high-place
- **Points:** Identifies the riders with more races
- **Unweighted:** Identifies consistent stage winners (general classification results)
- **Time Difference:** Identifies riders with large winning margins (even in fewer races)
- **Normalized Time Difference:** Improve fairness compared to Time Difference, since it normalizes stage length

This work analyses the history of Tour de France results, offering a broad view of rider performance across different eras. We introduced a new method for comparing riders using stage times and rankings. The use of multiple edge-weighting strategies provided a more deeper understanding of rider strengths. PageRank helped rank riders beyond just stage wins, while the Leiden algorithm revealed generational structures and clusters.

The general classification standings we used to create networks don't provide the whole view. By using diverse performance metrics and network modelling, we got patterns missing in more traditional rankings. With this, we can not only find the best riders by winning tours, but also by consistence performance and other metrics. Network analysis offers a more complex framework for the evaluation of successful professional riders in Tour de France.

In the future, we could explore temporal networks to track careers and performance. A deeper analysis could also reveal the best rider for specific stage types like time trials or summit finishes. To achieve this we could only look at important stages for each type. Additionally, we could use some interactive visualization tools to allow better insight into different rankings. Finally, applying this methodology to other Grand Tours like the Giro d'Italia or Vuelta a España could offer valuable comparative insights.

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